

Curriculum Vitae

Dr. Kalpana Chauhan, SMIEEE

Ph.D. (IIT Roorkee), PostDoc (University of Texas at Austin, USA)

Assistant Dean Student Welfare, Department IQAC Coordinator

Assistant Professor, Department of Electrical Engineering, SOET

IEEE Student Branch Counselor, IAS/Sensor Council Chapter and WIE Affinity Group Advisor

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PROFESSIONAL EXPERIENCE (13 Year Plus)

Name Of The Organization	Designation	Period	Tasks
Central University of Haryana	Assistant Professor	Jan, 16 2020-till date	Teaching
Galgotias College of Engineering and Technology, Greater Noida, India	Associate Professor	July 31, 2017 to Jan, 14 2020	Teaching and Research
SIRDA Group of Institutions Sundernagar, Himachal Pradesh, India	Assistant Professor	January 2013-May 2017	Teaching and Research
Indian Institute of Technology Roorkee, India	Ph.D. Scholar Cum Teaching Assistant	August 2009- January 2013	Teaching and Research

ACADEMIC QUALIFICATION

Course	Discipline	Board/University	Year	CGPA/ %
Ph.D.	Electrical Engineering	IIT Roorkee	2014	8.2
M.Tech	Control and Instrumentation Engineering	N.I.T. Jalandhar	2009	8.55
B.Tech	Electrical and Electronics Engineering	COER Roorkee	2006	67.78%
12 th	PCM	U.P.Board Allahabad	2001	65%
10 th	PCMB	U.P.Board Allahabad	1999	73.83%

RESEARCH INTEREST

- Electric Vehicle infrastructure and BTMS
- Medical Image Processing
- Monitoring and Control of Energy Storage System

SUBJECTs TAUGHT

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| <ul style="list-style-type: none"> ❖ Signals and Systems ❖ Control Systems ❖ Power Electronics ❖ Switch gear and Protection | <ul style="list-style-type: none"> ❖ Power Transmission and Distribution ❖ Digital Signal Processing ❖ Basic Electrical Engineering ❖ Non-Linear and Optimal Control |
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SPONSORED PROJECTS				
S.No.	Title	PI/Co-PI	Funding Agency	Ongoing/ Completed
1.	Solar based Charging Station of Electric Vehicles at CUH Campus (10,00000)	PI	UGC	Completed
2.	Solar based cold storage for Indian farmers (25000)	PI	IEI	Completed

FELLOWSHIP/ AWARDS/GRANTS
1. Guest of Honour, “ATAL Faculty Development Programme on "Renewable Energy Resources and Rural Electrification: Challenges and Solutions" sponsored by AICTE Training and Learning (ATAL) Academy at Dayalbagh Educational Institute (DEI), Agra, 10-21 Oct 2022. 2. Award of Honour, The Institution of Engineers (India), Shimla Section 2019-20. 3. Young Engineer Award (Electrical Engineering), The Institution of Engineers (India) 2018-19. 4. Awarded with Women of Science (WOS-A) Scheme, Department of Science and Technology, Government of India, 2018. 5. Best Paper Award on Paper entitled, "Segmentation of Mitral Regurgitant Jet using the Combination of Wavelet and Watershed Transformation,” in 8th IEEE Colloquium on Signal Processing & Its Applications (CSPA 2012), held in Melaka Malaysia during 23 to 25 March 2012. 6. Best Paper Award on Paper entitled “Statistical Analysis of speckle noise reduction techniques for echocardiographic Images,” International Conference on Methods and Models in Science and Technology (ICM2ST-11), held in Jaipur in Nov 2011. 7. MHRD Fellowship for Ph.D. research. 2009. 8. MHRD Fellowship for M. Tech research. 2007. 9. Graduate Aptitude Test in Engineering (GATE), 2007. Awards for my Students: 1. First Prize for “Design of Droop Control Based Real-Time Battery Management System for Automated DC Microgrid,” IEEE IAS CMD Myron Zucker Undergraduate Student Design Contest 2019, Baltimore, Maryland, United States. (Cash Prize: \$1300+ Full funding to Attend IAS Annual Meeting in the USA) 2. Solar based Hybrid Cold Storage is funded by "The Institution of Engineers (India)" under the R & D grant in aid scheme. Project Group: [UG students: Aniruddh Ashish, Aneeh Verma Guided by Dr. Kalpana Chauhan, Dr. Rajeev Kumar Chauhan]

Expert Talks
1. Expert talk on “Leveraging AI/ML for Smarter Energy System” in National Seminar “Machine Learning and AI using MATLAB”, organised by Electrical Department, UIET MDU Rohtak, 8th Sep 2025. 2. Expert talk on "Optimizing EV Charging with AI: Efficiency, Security, and Scalability" during our FDP "Digital Transformation in Power System: AI, Blockchain & Cybersecurity (DTPS-2025)" jointly organized by NIT Patna and Dayalbagh Educational Institute, Agra 3. Expert talk on “Synergizing BMS and EV Charging Infrastructure for Scalable E-Mobility” in Online STTP/FDP on Industrial Drives and EV Systems Department of Electrical Engineering, National Institute of Technical Teachers' Training & Research (NITTTR), Kolkata, 7th-11th July 2025. 4. Expert Speaker, “TWO DAYS NATIONAL WORKSHOP on DESIGN & EXPERIMENTAL DEMONSTRATION of ELECTRONICS CIRCUITS” organised by BOSE SOCIETY, DEPARTMENT OF PHYSICS RAO PAHLAD SINGH DEGREE COLLEGE, MAHENDERGARH IQAC, RPSDC, 27-28 Feb 2025. 5. Expert talk on Expert talk on "EV Charging: Strategy, Solutions and Opportunities" in One Week Short Term Course (STC) on “Advanced Control, and Machine Learning Approaches for Smart Distribution Systems (STC-2024)” at NIT Meghalaya, February, 03-07 2024. 6. Expert talk on “EV Charging: Strategy, Solutions and Opportunities” in One Week STTP on "Recent Research Trends in Electrical Engineering” REC Bijnor, September 11-15, 2023 7. Expert talk on “Modern Methods and Tools for Critical Disease” in 6 days online SEMINAR on "Artificial Intelligence Techniques for Healthcare Applications (AITHA-2023)" sponsored by SERB at NIT Delhi, 22Feb.-2023 to 27th Feb.-2023. 8. Expert talk on “Emerging Technologies for EV Charging with green infrastructure” in “ATAL Faculty Development Programme on "Renewable Energy Resources and Rural Electrification: Challenges and

Solutions" sponsored by AICTE Training and Learning (ATAL) Academy at Dayalbagh Educational Institute (DEI), Agra, 10-21Oct 2022.			
9. Expert talk on “Global Electrification: EV and Charging Station” in eight-day online faculty development program (FDP) on “Role of Power Electronics & Advances in Infrastructure for E-Mobility in India” during July 11-19, 2022, jointly organized by the Department of Electrical Engineering, NIT Warangal and Department of Electrical Engineering, MNIT Jaipur.			
10. Expert talk on “Automated Diagnosis of Cardiac Diseases for Remote Areas” in One week FDP on “Advanced research trends in signal and image processing” (Hybrid mode) at Jaypee Institute Noida, July 28–August 03, 2022.			
11. Expert talk on “Role of Electric Vehicles for Rural and Urban Development” in ATAL FDP on “Microgrid an Opportunity: Electric Vehicles and Renewable Energy Resources” sponsored by AICTE Training and Learning (ATAL) Academy at Dayalbagh Educational Institute (DEI), Agra, 5-9 July 2021.			
12. Expert talk on “Approachable Diagnosis Methods and Tools for Critical Diseases: Rural and Urban Areas”in IAS Chapter virtual meeting at IEEE Student Branch College of Engineering Karunagapally, 20 June 2021.			
13. IAS YP Lecture on “Approachable Diagnosis Methods and Tools for Critical Diseases: Rural and Urban Areas”in IAS CMD Lecture Series On Engineering & Methods And Tools Virtual Chapters Meeting Program, 29 Oct 2020.			
14. Expert talk One week FDP on Advanced research trends in signal and image processing (Hybrid mode) organised during (July 28–August 03, 2022).			
COMPUTER SKILL			
❖ MATLAB ❖ SCADA, PLC ❖ LAB VIEW		❖ C, C++ ❖ SQL	
TRAINING			
Name Of The Industry	Designation	Period	Nature Of Work
Bharat Heavy Electrical Limited, Haridwar, India	Trainee Engineer	May 16- June 29, 2005	CNC (Computer Numerical Controlled) Machine System
THESIS / DISSERTATION/ PROJECT			
Title		Guide	Degree
Processing of Echocardiographic Images with Reference to Mitral Regurgitation		Prof. M. L Dewal	Ph.D.
Application of HMI in Process Control		Dr. Rajesh Singla	M.Tech
Designing of an EEG signal measurement Device		Dr. S. K. Pahuja	M.Tech
Designing of Two levels Electronic Security and Safety system		Dr. M. A. Ansari	B.Tech
MEMBERSHIP OF PROFESSIONAL SOCIETIES			
❖ Senior Member, Institute of Electrical and Electronics Engineers (IEEE) ❖ Senior Member, Women in Engineering (WIE) ❖ Senior Member, Industry Applications Society (IAS) ❖ Member, Institute of Doctors, Engineers and Scientists (IDES) ❖ Member, International Association of Computer Science and Information Technology, Singapore. ❖ Member, International Association of Engineers (IAENG) Hong Kong. ❖ Full Member, Organization for Women in Science for the Developing World (OWSD) ❖ Member, The Institution Engineers (India)			
EDITOR/REVIEW BOARD/CONFERENCE COMMITTEES/SESSION CHAIR			

Editorial Board

1. Editorial Board Member, Journal Publication of International Research for Engineering & Management.
2. Editorial Board Member, International Journal of Cardiology and Cardiovascular Medicine.
3. Editor, International Journal of Advanced Computer Science and Information Technology, Switzerland.

Reviewer

1. IET Book Proposal-02
2. Journal of Elsevier.
3. IEEE Transactions and Journals.
4. Journals of Springer.
5. Journals of Nature
6. IEEE PES Innovative Smart Grid Technologies (ISGT) Middle East, 23-26th November 2025, Dubai, UAE
7. International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI-2025)” organised by Gautam Buddha University Greater Noida, 16-18 Jan 2025.
8. International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI-2026)” organised by Gautam Buddha University Greater Noida, 12-14 Feb 2026.
9. ICSC 2023, Department of Electronics and Communication Engineering, Jaypee Institute of Information Technology, A-10, Sector 62, Noida, U.P. 201309 (India).
10. IEEE 5th International Conference on Sustainable Energy and Future Electric Transportation (SEFET 2025).

Conference Members:

1. **Member of Scientific Committee**, International Engineering and Natural Science Conference (IENSC2018), on November 14-17, 2018 in Diyarbakir, Turkey.
2. **Technical Chair**, International Conference on Innovation and Application in Science and Technology (ICIAST-2021), Greater Noida, India, 21-23 December 2021
3. **Member of Panel Discussion**, “3rd International Symposium on Sustainable Energy and Technological Advancements (ISSETA 2024)” organized by National Institute of Technology Meghalaya, Shillong, India during 23rd-24th February 2024.
4. **Session chair** in the 1st International Conference on Smart and Sustainable Technologies in Energy and Power Sectors (SSTEPS-2022), organized by Central University of Haryana, Haryana, India, during November 7– 9, 2022.
5. **TPC Member** ICSC 2023, Department of Electronics and Communication Engineering, Jaypee Institute of Information Technology, A-10, Sector 62, Noida, U.P. 201309 (India).
6. **Session Chair** IEEE PES Innovative Smart Grid Technologies (ISGT) Middle East, 23-26th November 2025, Dubai, UAE
7. **Session Chair** International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI-2025)” organised by Gautam Buddha University Greater Noida, 16-18 Jan 2025.
8. **Session Chair** International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI-2026)” organised by Gautam Buddha University Greater Noida, 12-14 Feb 2026.

Short Term/Workshops Organised

1. Webinar on “Microgrids: Enabling a Smarter Energy Future” 27 April 2026.
2. Workshop on “Awareness About Virtual Labs”, 21 Nov 2025.
3. Dohan Food and Cultural Festival, 12/11/2025.
4. Vigilance Awareness Week 2025, “Say No to Corruption; Commit to the Nation”, 31 Oct 2025.
5. IEEE Workshop on Electric Vehicle System Level Modelling 18th to 22nd Aug 2025.
6. Online FDP on “Emerging Technologies in Engineering, Medical Instrumentation, EVs, and Microgrid Systems” Organized by Department of Electrical Engineering, 15th – 21st May 2025.
7. IEEE SB Officers Training, 28th March 20, 2025.
8. IEEE WIE Event on SHESPEAKS, the world listens Organized by IEEE SB at CUH, 06/05/2025.
9. Expert Lecture on “Power Quality Management in Smart Grids”, 3rd October 2024.
10. Expert Lecture on “Reliability Modelling and Evaluation of Engineering Systems, 12th Nov 2024.
11. Expert Talk on “Online talk on Entrepreneurship and Management, IEEE WIE SB AG, IEEE Student Branch, 13th Feb 2024.
12. Expert talk on “Tailoring Your Resume to Every Opportunity”, IEEE WIE SB AG, IEEE Student Branch, 11th Apr2024.

	13. ATAL FDP on “AICTE training and learning (ATAL) academy Faculty Development Programme (FDP) on “Microgrid with Electric Vehicles: Intelligent and Allied Areas”, 23rd – 28th October 2023. 14. IEEE Day Celebration, 22 November 2023. 15. ABHIYANTA 2.0, Engineers Day Celebration, 15 November 2023. 16. Workshop on Virtual Lab: Control Engineering 7 Dec 2022. 17. One Day Hands on workshop on MATLAB and IoT, IEEE SB at CUH, 26 Aug 2022. 18. IEEE Webinar on Solid-state energy conversion and signal modulation using Ferroelectrics organized by IEEE Student Branches at CUH, 16th January 2022. 19. IEEE Workshop on Modeling and Simulation with MATLAB for Emerging Applications Jointly organized by IEEE SB DEI Agra, IEEE SB CUH Mahendragarh, and Mathworks 18-28 October 2021. 20. Engineer’s Day Celebration 15 Sep 2021. 21. Expert lecture on “Making IEEE your Professional Home” 5-Sep-2021 22. International webinar on the Fingerprint 06-Dec-2020 1 day IEEE 23. International webinar on Vocal 4 Local summit 2020 20-Nov-2020 1 day IEEE 24. International Webinar on Haryana Technology Conclave 16-Nov-2020 1 day IEEE 25. Webinar on Engineers: Ideas, Innovation and creativity 15-Sep-2020 1 day IEEE 26. Inauguration ceremony of IEEE SB Central University of Haryana 24-Aug-2020 1 day IEEE 27. Webinar on, Power of Professional Networking & Volunteering, Central University of Haryana, 09-Jun-2020 1 day IEEE 28. Webinar on Smart DC Microgrid Solutions for Rural Areas Electrification 28-May-2020 1 day IEEE 29. Webinar on DCS and Its Applications to Power Sector 25-May-2020 1 day IEEE 30. Technical workshop cum Training on Robotics at SIRDA Group of Institutions 13-Apr-2013 to 14-Apr-2013 2 days.
Short Term/Workshops Attended	
1.	Seven days Faculty development program, Amity university Chhattisgarh on कणाद से रमन- Mainstreaming Indian Knowledge System (IKS) in Higher Education, organized by Amity Institute of Biotechnology, Amity University Chhattisgarh, Raipur, Chhattisgarh held on 16th – 22nd July 2025.
2.	NEP 2020 Orientation & Sensitization Programme on Research and Development, organized by MMTTC at the Central University of Haryana, Mahendergarh (India) from November 20 to 30, 2023.
3.	Two - week Interdisciplinary Refresher Course in “Advanced Research Methodology” from 23 January - 06 February, 2023.
4.	Faculty Induction Program “Faculty in Universities/Colleges/Institutes of Higher Education”, from 20 April – 19 May, 2022.
5.	ATAL Faculty Development Program on “Microgrid an Opportunity: Electric Vehicles and Renewable Energy Resources” from 2021-07-05 to 2021-07-09 at Dayalbagh Educational Institute, Agra.
6.	Faculty Development Program on “Renewable Power Generation, Control and Grid Integration”, Indira Gandhi Institute Of Technology, Sarang, 10-14 Aug 2020
7.	Short Term Course on "Recent Trends in Power System Planning, Monitoring and Control" NIST Brahmipur, 10-12 Aug 2020
8.	Faculty Development Programme on “Development and Delivery of MOOCs and E-Content” , June 26-July 1, 2020
9.	International short term course on “Microgrid Opportunity: Renewable Energy Resources and Buildings” June 16-20, 2020.
10.	Webinar on "Leveraging Clean Energy and Humanitarian Technologies” , June 2, 2020.
11.	Webinar series on smart grid, project Management , online learning, and application of engineering in practical life, May 28-31 2020.
12.	Webinar on “certified virtual classroom teacher” May 23 2020
13.	STTP on “Indian Electricity rule and code practices” Dec 16-20, 2019.
14.	Short Term Course on “Optical Fibers: Potential and Applications through ICT” Conducted by NITTR Chandigarh, at Galgotias College of Engineering and Technology Greater Noida, 18-22 December 2017.
15.	QIP workshop on “Breathing Mechanics” organized by the Department of Mathematics under QIP program at IIT Roorkee, March 31, 2013

16.	QIP workshop on "Interaction of Academic-Industry on Mathematical Modeling," organized by the Department of Mathematics under QIP Program at IIT Roorkee, March 31, 2012.
17.	NI-LabVIEW- Basics workshop held on 25 march 2012.
18.	"Library oriented program and workshop on Scopus and science Direct" organized by the Mahatma Gandhi central Library, IIT Roorkee, February 27, 2012.
19.	Workshop on " Web of Science (Science Citation Index) & SciFinder Scholar (Chemical Abstract))" Organized by Mahatma Gandhi central Library, IIT Roorkee, June 23, 2011.
20.	Training Program on "Ultrasound Imaging, Processing and Clinical Applications" organized by Department of Electrical Engineering, IIT Roorkee, 27-29 Jan2010.
Patents	
1.	Kalpana Chauhan, Shaurya Chauhan, Rajeev Kumar Chauhan, "Blower Drone for Cleaning of the Cricket Grounds", 12/02/2025, (Application number: 202511011827).
1.	Kalpana Chauhan, Rajeev Kumar Chauhan, Adarsh Tiwari, Bhukya Vamshi, "Smart cleaning and avian deterrent system for solar panels" 24/01/2024, (Application number: 405498-001) Granted
2.	R.K. Chauhan and K. Chauhan, "Design and method of priority based multi-level charger for electric vehicles ",filed 28/0923, Published 20 Oct 2023, (Application Number: 202311065104)
3.	K. Chauhan, and R.K. Chauhan, "Three-Port Grid-tied Solar EV Charger", 4July 2023, (Application Number: 389594-001) Granted
4.	R. K. Chauhan, and K. Chauhan, "System design for automatic vehicle theft detection at the toll plazas," filled on 12 July 2022 (Published) (Application Number: 202211039883)
5.	K. Chauhan, and R.K. Chauhan, System design for echocardiography machines to find out the vena contracta width automatically for the mitral regurgitation assessment, filed on March 22, 2019. (Published) (Application Number: 201911011079), Granted 523243
6.	R.K. Chauhan, and K. Chauhan, Method and system for rechargeable battery monitoring in photovoltaic based microgrid system, filed on February 18, 2019. (Filed and published) (Application Number: 201911006250), Granted 538360
7.	K. Chauhan, and R.K. Chauhan, System for echocardiography machines to find out jet area automatically for the mitral regurgitation severity evaluation, filed on November 22, 2018. (Granted: 29/02/2024) (Application Number: 201811043976),Granted.
8.	R.K. Chauhan, and K. Chauhan, Energy management system for photovoltaic-battery based DC microgrid in buildings, filed on July 17, 2018. (Granted: 12/02/2024) (Application Number: 201811026751), Granted (509563).
Books	
1.	R.K. Chauhan, and K. Chauhan, "Distributed Energy Resources in Microgrids" Elsevier, August 01, 2019 (ISSN: 9780128177747)
2.	R.K. Chauhan, K. Chauhan, and S.N. Singh,"Microgrids for Rural Areas: Research and case studies" August 01, 2020, IET (ISBN: 978-1-78561-999-1)
3.	K. Chauhan and R.K. Chauhan, "Image Processing Methods for Automatic Diagnosis of Cardiac Diseases", Elsevier, July 13, 2021 (ISBN: 9780323850643)
Publications in the International Journals	
1.	Kalpana Chauhan, Rajeev Kumar Chauhan, "Thermal Runaway Prediction and Risk Assessment in Lithium-Ion Batteries using Physics-Informed Neural Network", Scientific Reports, Nature, SCI:4.5,Submitted.

2.	Kalpna Chauhan, Rajeev Kumar Chauhan, "Physics-Informed Neural Networks for Quantitative Assessment of Mitral Regurgitation", Biomedical Signal Processing and Control, Elsevier, SCI:4.9, Submitted.
3.	Rajat Yadav, Kalpna Chauhan, Rajeev Kumar Chauhan, CHARGER FAULT RATE ANALYSIS-BASED OPTIMAL ELECTRIC VEHICLE CHARGING STATION SCHEDULING AND ROUTING FRAMEWORK, "Journal of Energy Storage" Elsevier, Accepted SCI:9.8
4.	Poonam Sharma, Rajesh Kumar Dubey, Kalpna Chauhan, "A REVIEW ON OPTIMAL CONTROL STRATEGIES FOR CHARGING AND DISCHARGING OF ELECTRIC VEHICLE BATTERY", Journal of Advanced Energy Research , Bentham Science, Submitted.
5.	Poonam Sharma, Rajesh Kumar Dubey, Kalpna Chauhan, "A Bi-Directional Phase-Synchronized V2G Frequency Regulation Using FACGMLC and FCAPID Controllers for THD Reduction", EPSR, Elsevier, Submitted.
6.	Poonam Sharma, Rajesh Kumar Dubey, Kalpna Chauhan, "A MASVRL AND AQNFSIS-BASED OPTIMAL CONTROL STRATEGIES FOR V2G IN TERMS OF LOSS REDUCTION", Discover Electronics, Springer, Submitted.
7.	Chauhan, K., Chauhan, R.K. Machine Learning-Based Real-Time Dynamic Programming for Battery Thermal Management System (ML-RTDP-BTMS) for Electric Vehicles. Iran J Sci Technol Trans Electr Eng (2026). https://doi.org/10.1007/s40998-026-01032-3 . SCI:1.4
8.	Yadav, R., Chauhan, K., & Chauhan, R. K. (2024). An Effective Requests-Based Charging Station Scheduling and Routing Model with HUI-MTA ANN and BD-SSO for EV Charging. International Journal of Energy and Environmental Engineering, 15(02 (June 2024)). https://doi.org/10.57647/ijeec.2024.1502.08 SCI: 2.4
9.	V. Amoriya, R. K. Chauhan, S. Mittal, S. Sharma, V. Panit, and K. Chauhan, "Modelling of Multi-Phase Interleaved based Universal Charging Port for Electric Vehicles," Smart Grids and Sustainable Energy, Vol. 9, pp. 1-16, 2024. SCI:1.9
10.	V. Amoriya, R. K. Chauhan, V. Panit, Jahanvi, S. Verma, S. Mittal, K. Chauhan, S. Upadhyay, and A. Cardena, "Modeling of Bidirectional Electric Vehicle Charger for Grid Ancillary Services," International Journal of Emerging Electric Power Systems, <i>International Journal of Emerging Electric Power Systems</i> , vol. 25, no. 1, 2024, pp. 61-74 DE GRUYTER, 2023. (ISSN:1553-779X) (Web of Science – Emerging Sources Citation Index) https://doi.org/10.1515/ijeeps-2022-0234 . SCI:0.7
11.	R. K. Chauhan, and K. Chauhan, "Intelligent toll collection system for moving vehicles in India," <i>Intelligent System with Applications</i> , Elsevier, September 2022. Vol. 15, pp. 200099. (SCI-4.3) doi: https://doi.org/10.1016/j.iswa.2022.200099 . SCI:4.3
12.	R. K. Chauhan, K. Chauhan, and A.Q.H. Badar "Optimization of Electrical Energy Waste in House Using Smart Appliances Management System-A Case Study," <i>Journal of Building Engineering</i> , Elsevier, vol. 46, 1 April 2022. DOI: https://doi.org/10.1016/j.jobbe.2021.103595 (ISSN: 2352-7102) (SCI Index: Impact Factor 7.4).
13.	R. K. Chauhan, and K. Chauhan, "Smart Protection System for Identification and Localization of Faults in Multi-Terminal DC Microgrid," IET Smart Grid Vol. 3 pp. 882-889, 2020.ISSN: 2515-2947. 10.1049/iet-stg.2019.0315 (SCI:2.7)
14.	R.K. Chauhan, K. Chauhan, "Battery Monitoring and Control System for Photovoltaic based DC Microgrid", International Journal of Emerging Electric Power Systems, vol 20, no. 6. 2019. ISSN: 1553-779X, https://doi.org/10.1515/ijeeps-2019-0133 . SCI:0.7
15.	R.K. Chauhan, K. Chauhan, B. R. Subrahmanyam, A. G. Singh, and M. M. Garg, "Distributed and Centralized Autonomous DC Microgrid for Residential Buildings: A Case Study," <i>Journal of Building Engineering</i> , Elsevier, vol. 27, pp. 1-6, January 2020 (ISSN: 2352-7102) , (SCI Index: Impact Factor 7.4) https://doi.org/10.1016/j.jobbe.2019.100978
16.	K. Chauhan, R. K. Chauhan, "Design and Development of Two Levels Electronic Security and Safety System for Buildings", <i>International Journal of Electronic Security and Digital Forensics</i> , Inderscience, vol. 12, no. 3, pp. 279-292, 2020, ISSN: 1751-9128. https://doi.org/10.1504/IJESDF.2020.108297 (SCI: 0.5)

17.	R. K. Chauhan and K. Chauhan, "Building Automation System for Grid-Connected Home to Optimize Energy Consumption and Electricity Bill," <i>Journal of Building Engineering, Elsevier</i> , vol. 21, pp.409-420, January, 2019. https://doi.org/10.1016/j.jobbe.2018.10.032 (ISSN:2352-7102) (SCI: Impact Factor 7.4)
18.	R. K. Chauhan, K. Chauhan, and Josep M Guerrero, "Controller design and voltage stability analysis of grid connected DC microgrid," <i>Journal of Renewable and Sustainable Energy</i> , vol.10, no.3, pp. 035101-10, April 2018. (ISSN:1941-7012) https://doi.org/10.1063/1.5024714 , (SCI: Impact Factor 2.3)
19.	R. K. Chauhan and K. Chauhan, "Performance Analysis of Power Distribution Systems with Weakly Grid Connected Rural Homes in Indian," <i>Energy and Buildings, ELSEVIER</i> , vol. 172, pp. 307-316, Aug. 01, 2018. https://doi.org/10.1016/j.enbuild.2018.05.003 , ISSN: 0378-7788 (SCI: Impact Factor 7.1) ,
20.	K. Chauhan and R. K. Chauhan, "Optimization of grid energy using demand and source side management for DC microgrid," <i>Journal of Renewable and Sustainable Energy</i> , vol. 9, no. 3, pp. 035101, 2017. (DOI: http://dx.doi.org/10.1063/1.4984619).(ISSN:1941-7012) (SCI: Impact Factor 2.3)
21.	K. Saini, M. L. Dewal, and M. K. Rohit, "A fast region-based active contour model for boundary detection of echocardiographic images," <i>Journal of Digital Imaging, Springer</i> , vol. 25, no. 2, pp.271-278, 2012. (ISSN: 0897-1889). 10.1007/s10278-011-9408-8 . (SCI: Impact Factor 3.8)
22.	M. L. Dewal, K. Saini, and M. K. Rohit, "Assessment of mitral regurgitation severity with intensity-based region growing," <i>International Journal of Hybrid Information Technology (IJHIT)</i> , vol. 8, no. 6, pp. 45-56, 2015. ISSN:1738-9968. http://dx.doi.org/10.14257/ijhit.2015.8.6.05 (Scopus Index) .
23.	K. Saini, M. L. Dewal, and M. K. Rohit, "Detection of mitral regurgitation severity using color histogram," <i>International Journal of Research in Biomedicine and Biotechnology, URP Journals</i> , vol. 3, no. 2, pp.102-108, June 2013. Peer reviewed
24.	K. Saini, M. L. Dewal, and M. K. Rohit, "A mathematical evaluation of mitral regurgitation severity with EROA," <i>International Journal of Cardiovascular and Cerebrovascular Disease</i> , (ISSN: 2331-9887), vol. 1, no. 1, pp.10-14, 2013., 10.13189/ijccd.2013.010102 .ISSN: 2331-9895. peer-reviewed .
25.	K. Saini, M. L. Dewal, and M. K. Rohit, "Automatic detection of vena contracta width for the mitral regurgitation assessment," <i>International Journal of Computer Science and Management Research</i> , vol. 2, no. 5, pp. 2455–2460, May 2013 (ISSN: 2278-733X) (Impact Factor 2.11) .
26.	K. Saini, M. L. Dewal, and M. K. Rohit "Level set based on new signed pressure force function for echocardiographic image segmentation," <i>International Journal of Innovation and Applied Studies</i> , vol. 3, no. 2, pp. 560-569, June, 2013, Peer reviewed
27.	K. Saini, M. L. Dewal, and M. K. Rohit, "Automatic jet area detection during mitral regurgitation," <i>International Review on Computers and Software (IRECOS)</i> , Praise worthy Prize, vol.7 no. 6, pp. 2891-2898. ISSN: 1828-6003, (Index in ICI Journal Master List), 2013 (Scopus Index)
28.	K. Saini, M. L. Dewal, and M. K. Rohit, "Modified nonlinear complex diffusion filter (MNCDF)", <i>Journal of Echocardiography, Springer</i> , vol. 10, pp. 48-55, 2012. ISSN: 1349-0222. (SCI IMPACT FACTOR 1.8). DOI: 10.1007/s12574-012-0119-z
29.	K. Saini, M. L. Dewal, and M. K. Rohit, "A modified hybrid filter for echocardiographic image noise removal," <i>International Journal of Signal Processing, Image Processing and Pattern Recognition</i> , pp. 61-72, vol. 5, no. 2, 2012. (ISSN: 2005-4254).
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Thesis/ Projects Supervised		
S.No	Degree	Nos.
1.	PhD	02
2	M.tech	05
3	B.Tech	25
Outreach/Administrative Activities		
❖ Assistant Dean Student Welfare, Central University of Haryana. ❖ Department IQAC Coordinator, Central University of Haryana. ❖ Convenor of Socio-Economically Disadvantaged Groups Cell (SEDG). ❖ Nodal coordinator, Virtual Labs at Central University of Haryana. ❖ Member of TechFest 2023 at Central University of Haryana. ❖ Member of NEP Task Force for Central University of Haryana. ❖ Counselor, IEEE Student branch, Central University of Haryana. ❖ Advisor, IEEE IAS Chapter, Central University of Haryana. ❖ Advisor, IEEE Sensor Council, Central University of Haryana. ❖ Counselor/ Advisor WIE IEEE Central University of Haryana. ❖ Editor Department Magazine (Electrical Department, Galgotias College of Engineering and Technology Greater Noida) ❖ Editor of the conference proceedings for International Conference on “Advance Computing, Communication, Energy, Instrumentation and Robotics (ICAC2EIR18)” will be held during 23-24 March 2018 at Galgotias College of Engineering and Technology Greater Noida. ❖ NBA Coordinator. ❖ Faculty Advisor International Conference on “Advance Computing, Communication, Energy, Instrumentation and Robotics (ICAC2EIR18) ❖ President of PhD Research Scholars, Department of Electrical Engineering, IIT Roorkee (2010-2012) ❖ Secretary of MRC married hostel IIT Roorkee (2011-2012) ❖ Proctor for Girls Department of Control and Instrumentation NIT Jalandhar (2008-2009)		

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